

Text Guided 3D Room Editor with Interactive Effects

Final Project Proposal

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1. Overall Goal

The goal of this project is to build an **interactive 3D indoor scene editing system** that allows users to (1) generate entirely new objects from text prompts using **Score Distillation Sampling (SDS)** and insert them into an existing **3D Gaussian Splatting (3DGS)** scene, and (2) **select existing objects in the scene** and **apply dynamic physical effects** such as melting, inflation, or jelly-like motion. The system targets **ScanNet indoor scenes** and integrates state-of-the-art components including 3DGS, SAM 2, Stable Diffusion, and PhysGaussian into a unified pipeline.

2. Expected Outcome

Upon completion, the system will support **two distinct editing modes**:

- **Mode A — New Object Generation**: Given a text prompt (e.g., “add a red sofa”), the system generates a new 3D Gaussian object via SDS and automatically places it at a physically plausible location in the scene (e.g., on the floor or on a table surface). The generated object can then be subjected to physical simulations such as melting or bouncing.
- **Mode B — Existing Object Editing**: Given a text query (e.g., “make the chair wobble like jelly”), the system uses SAM 2 to semantically segment and select the target object within the 3DGS scene and applies a user-specified physical effect using PhysGaussian.

The final deliverable will include a working code pipeline, quantitative evaluation results (PSNR, SSIM, LPIPS for scene quality; FPS for real-time performance), qualitative visual comparisons, and a written technical report.

3. Proposed Approach

The project is structured in four sequential steps:

Step 1 — Scene Reconstruction (Foundation)

3DGS is trained on ScanNet indoor scene data. Each Gaussian G_i is parameterized by position μ , covariance Σ , color c , and opacity α . The Gaussian function is defined as:

$$G(x) = \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

Parameters are optimized by minimizing a combination of L1 and D-SSIM losses against ground truth training images:

$$L = (1 - \lambda)L_1 + \lambda L_{D-SSIM}$$

The trained model serves as the base 3D scene representation for all downstream editing operations.

Step 2 — Semantic Segmentation with SAM 2

SAM 2 (Segment Anything Model 2) is used to identify and isolate target objects within the 3DGS scene given a text query. This replaces the LERF-based approach used in Feature Splatting (ECCV 2024), enabling a comparison between the two segmentation strategies in terms of accuracy and speed.

Step 3 — Generative Editing via SDS (Mode A)

For new object generation, the system initializes a set of random Gaussians in the masked region and optimizes their parameters using the Score Distillation Sampling (SDS) loss:

$$\nabla_{\theta} L_{SDS} = \mathbb{E}_{t, \epsilon} \left[w(t) (\epsilon_{\phi}(x_t; y, t) - \epsilon) \frac{\partial x}{\partial \theta} \right]$$

The 2D diffusion feedback from Stable Diffusion is backpropagated into the Gaussian parameters, progressively shaping the new object to match the text prompt. A Surface-Aware Placement module uses the depth information from surrounding Gaussians to automatically snap the generated object to the nearest physical surface (floor, table, shelf).

Step 4 — Physical Simulation with PhysGaussian (Modes A & B)

Before applying physics, the system must extract a physically recognizable surface from the Gaussian scene, since Gaussians are cloud-like and not directly interpretable by physics engines. This is done via two steps:

- **SDF (Signed Distance Field) Extraction:** The density information of the Gaussians is used to compute a mathematical boundary surface that defines where solid surfaces (e.g., floors, tables) exist in the scene.
- **Proxy Geometry:** A transparent collision box is placed over the background Gaussians so that the physics engine can compute collisions between inserted objects and the scene.

PhysGaussian then treats each Gaussian as a physical particle with material properties (mass, elasticity, viscosity). The Material Point Method (MPM) is used to simulate dynamic physical effects including melting (liquid material), inflation (volumetric expansion), and jelly-like oscillation (elastic material). This is applied to both newly generated objects (Mode A) and existing segmented objects (Mode B).

4. Key Contributions

#	Contribution	Novelty vs. Feature Splatting
1	SDS-based new object generation from text prompts	Feature Splatting cannot generate new objects; it only manipulates existing ones
2	SAM 2 replaces LERF for semantic segmentation	Comparison between two segmentation approaches in 3DGS editing
3	Surface-Aware Placement for auto-positioning	Novel module; not present in any prior 3DGS editing pipeline
4	Unified Mode A + Mode B pipeline	First system supporting both generative insertion and existing object physics in one framework

5. Deep Learning Topics

- 3D Gaussian Splatting (Novel View Synthesis, Differentiable Rendering)
- Score Distillation Sampling (Backpropagation through Diffusion Models)
- Segment Anything Model 2 (Vision Foundation Models, Zero-shot Segmentation)
- Physics-based Simulation with Neural Representations (PhysGaussian / MPM)
- Knowledge Distillation (potential future extension for model compression)

6. Third-Party APIs and Frameworks

Tool / Framework	Purpose	How Used
3D Gaussian Splatting (official repo)	Scene reconstruction	Train on ScanNet data to build the base 3D scene
SAM 2 (Meta AI)	Semantic segmentation	Segment and isolate target objects in the Gaussian scene via text query
Stable Diffusion (Hugging Face)	SDS-based object generation	Provide 2D diffusion feedback for generating new Gaussian objects
PhysGaussian	Physical simulation	Apply MPM-based material simulations (melting, jelly, inflation) to Gaussians

PyTorch	Model training & optimization	Core DL framework for all gradient-based computations
CUDA / cuDNN	GPU acceleration	Accelerate Gaussian rasterization and training

7. Dataset

- **ScanNet**: Indoor RGB-D scene dataset with camera poses. A subset of scenes will be selected for training and evaluation. ScanNet provides realistic indoor environments (living rooms, bedrooms, offices) well-suited for furniture-level object editing.

8. Resources

Research Papers

- Kerbl et al. (2023). 3D Gaussian Splatting for Real-Time Radiance Field Rendering. SIGGRAPH 2023.
- Wang et al. (2024). Feature Splatting: Language-Driven Physics-Based Scene Synthesis and Editing. ECCV 2024.
- Poole et al. (2022). DreamFusion: Text-to-3D using 2D Diffusion. ICLR 2023.
- Ravi et al. (2024). SAM 2: Segment Anything in Images and Videos. Meta AI.
- Xie et al. (2024). PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics. CVPR 2024.
- Dai et al. (2017). ScanNet: Richly-Annotated 3D Reconstructions of Indoor Scenes. CVPR 2017.

Textbook

- Prince, S. (2023). Understanding Deep Learning. Chapters 3–11 (Neural Networks, Training, CNNs, Generative Models).

9. Team Members and Work Distribution

This is a solo project. All work will be implemented and completed by a single team member.

10. Implementation Timeline

Week	Dates	Milestone	Priority
1–2	Mar 9 – Mar 22	Environment setup, ScanNet data preprocessing, 3DGS training (Step 1)	Must

3	Mar 23 – Mar 29	SAM 2 integration for semantic segmentation (Step 2)	Must
4–5	Mar 30 – Apr 12	SDS-based object generation + Surface-Aware Placement (Step 3, Mode A)	Must
5	Apr 13	Midpoint Evaluation: demo of Mode A (object generation)	Must
6	Apr 14 – Apr 20	SDF extraction + Proxy Geometry + PhysGaussian integration — melting, jelly, inflation (Mode A)	Must
7	Apr 21 – Apr 27	Mode B (existing object physics) + Shadow Mapping	Nice-to-have
8	Apr 28 – May 4	Quantitative evaluation, ablation study, final report, video tour	Must
After May 4	May 5–8	Written report, code cleanup, optional poster	Must

11. Scope and Risk Management

The project is designed with a clear Must-have / Nice-to-have split to ensure a successful submission even if unexpected implementation challenges arise:

- Must-have: Step 1 (3DGS scene reconstruction) + Step 2 (SAM 2 segmentation) + Step 3 Mode A (SDS generation + placement) + Step 4 SDF/Proxy Geometry + PhysGaussian effects (melting, jelly, inflation)
- Nice-to-have: Mode B (existing object physics), Shadow Mapping for realistic shadow casting, full ablation study (SAM 2 vs. LERF comparison)
- Future Work: Knowledge Distillation for model compression, large-scale urban scene extension

The primary technical risk is the integration of SDS with 3DGS, which requires careful alignment of the diffusion model's 2D feedback with the 3D Gaussian optimization loop. Two weeks are allocated to this step to allow sufficient debugging time.

Additionally, this project intentionally uses different models from those in Feature Splatting (e.g., SAM 2 for segmentation, Stable Diffusion for SDS-based generation, and PhysGaussian for physics simulation) and will conduct comparison experiments to evaluate performance differences. If any of these models do not yield sufficient results, alternative models may be explored to further enhance overall performance and robustness.
